

Fabrication techniques of Dental Ceramics

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Fabrication Techniques:

- One consequence of the strong ionic bonds in ceramics is that these materials melt only at relatively high temperatures.
- Some techniques are used to form the final dental restoration including stacking, infusing, pressing, or machining.

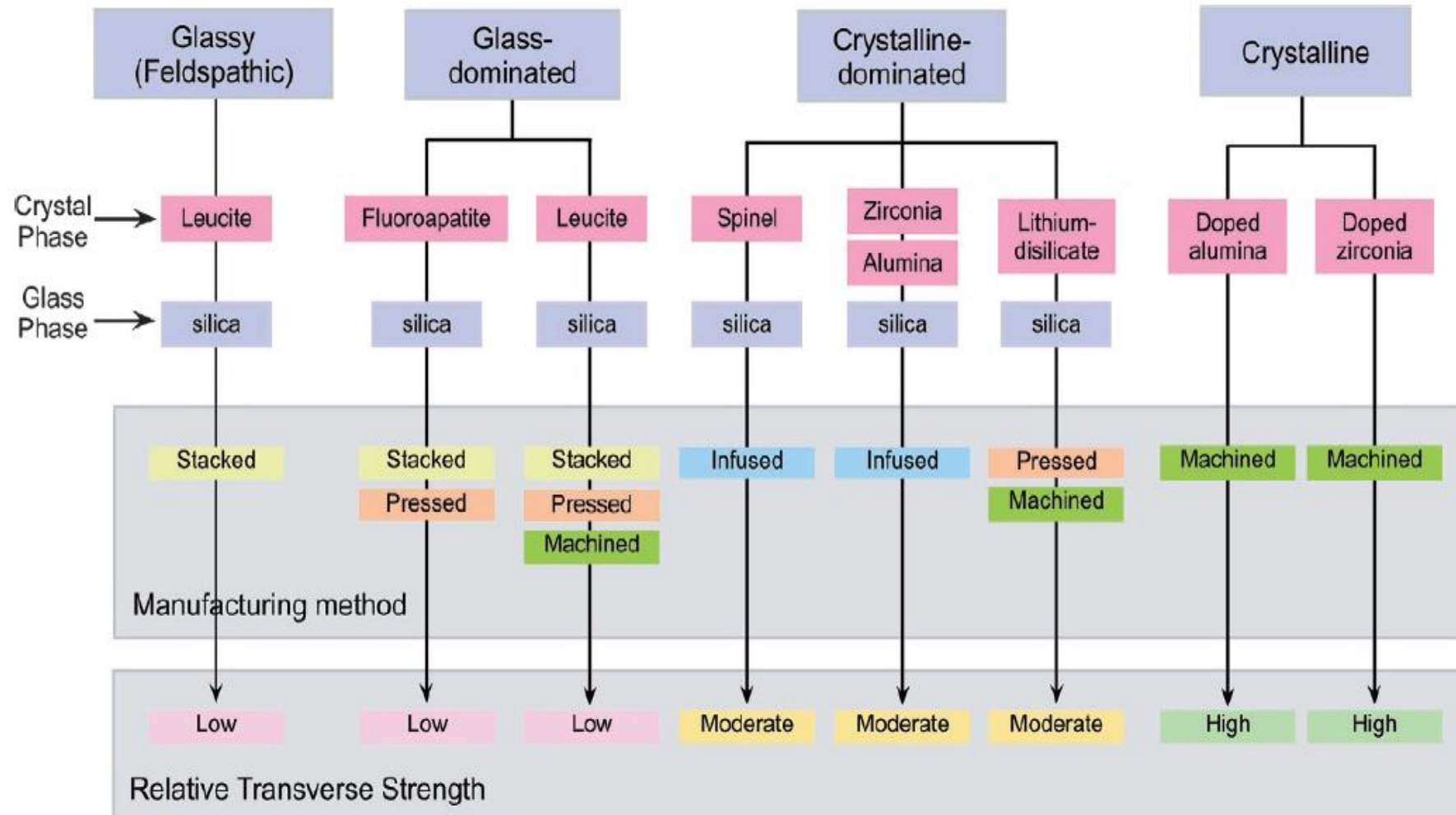


FIG 14-4 Summary of the major types of ceramics used in dentistry today; listing the identity of the crystalline phase, glass phase, common methods of manufacture, and relative transverse strength.

A-In stacking: the ceramic material is ground into particles that are manipulated into an approximation of the final form in a “green” state (weak state).

The green state is achieved by a process called condensation.



A viscous slurry of the porcelain particles in water or water-glycerol is applied to a substructure, **using vibration to pack the particles and expel the liquid to the surface.** Then liquid is removed by a sorbent tissue.

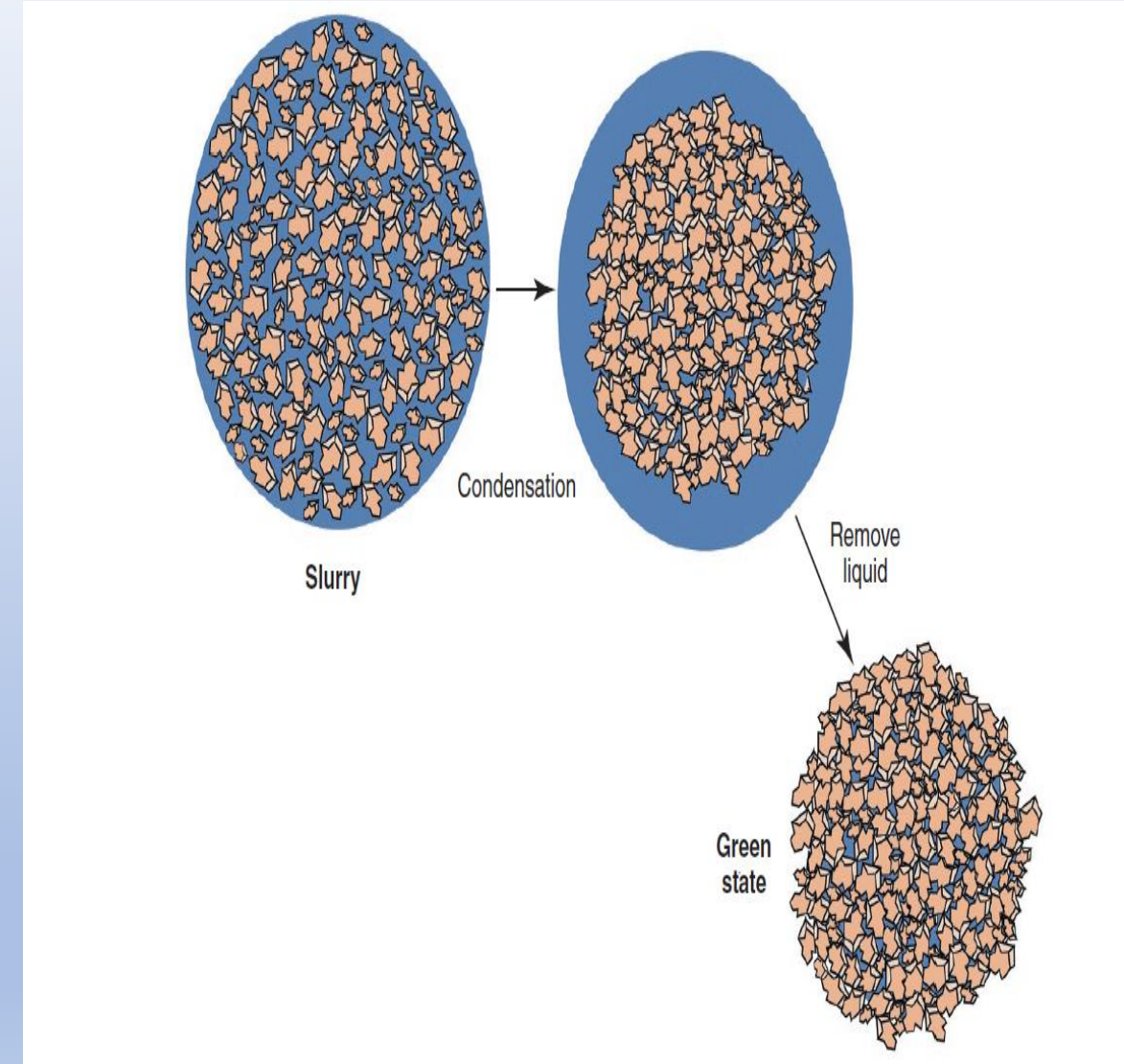


By alternatively vibrating the liquid to the surface and then removing it, the particles become packed quite tightly into the green state.

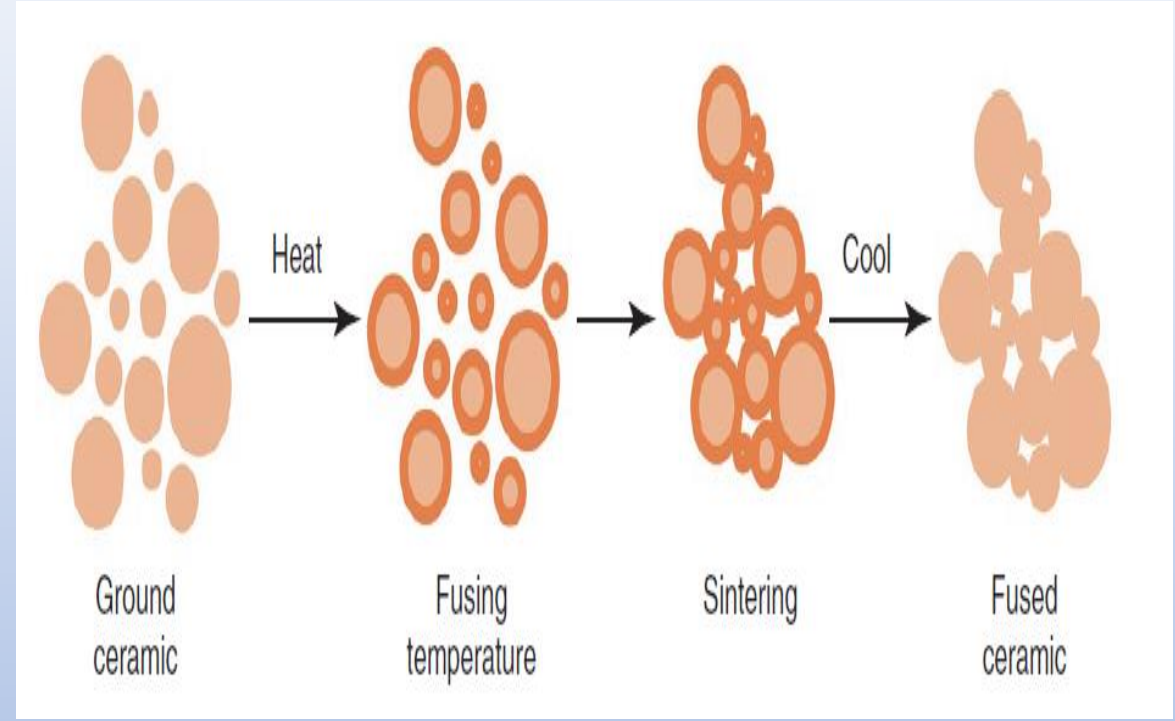
In the green state, the ceramic can be carved and shaped into approximate final form.

Applications: Stacking is used to 1) apply ceramics to alloys or high-strength ceramic substructures or 2) to fabricate laminate veneers.

Stacking is used primarily for glassy and glass-dominated ceramics



- B-Sintering:
- After forming the green state, the ceramic is fired in a specialized oven in a process called sintering.
- During sintering, the particles only melt and fuse at their surfaces.
- After a short time (minutes), the particles are cooled to room temperature, and the fired ceramic is formed.
- To reduce the “dead” air spaces in the ceramic, a vacuum is often applied in sintering.



- Important notes about sintering:
- The sintering temperature is often referred to as the fusing temperature of the ceramic.
- During sintering, significant shrinkage (up to 10% linearly) occurs, for this reason, the green state form is often made a bit larger than final desired contours.



- **C-Infusion:** is a fabrication method by which a porous sintered form is infiltrated with a silica glass (Figure 14-7).
- **In the initial step**, a ceramic slurry is stacked onto a model of a tooth preparation, sintered to provide an intermediate strength, and trimmed to near-final form
- **Next step**, a silica glass is added and the restoration is sintered again.
- During the second sintering, the silica melts and flows into all the capillary spaces of the initial structure.
- The infused ceramic has markedly higher strengths
- Infusion is used on a subset of crystalline-dominated ceramics



Reading only

Where are the capillary spaces for silica infiltration in second sintering?

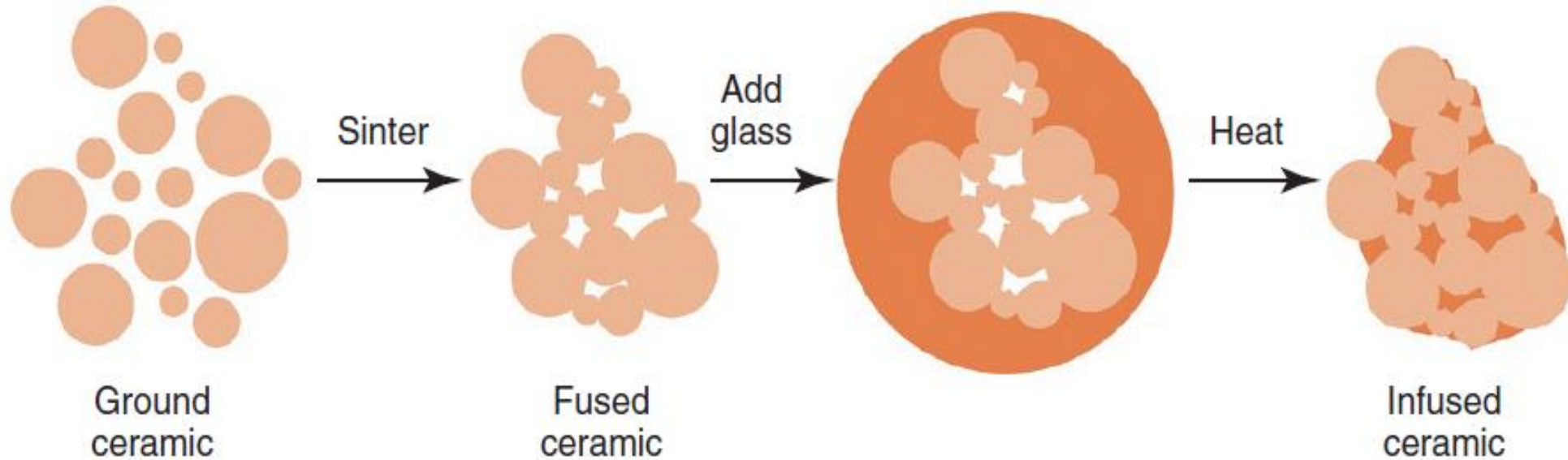


FIG 14-7 Infused ceramics. In some ceramics, a sintered crystalline structure (fused) is secondarily surrounded by a lower fusing (melting) glass to strengthen the ceramic. The amorphous glass, which is often silica, infiltrates throughout the ceramic to form an infused ceramic. The infused ceramic is capable of withstanding higher forces without fracture in many clinical applications.

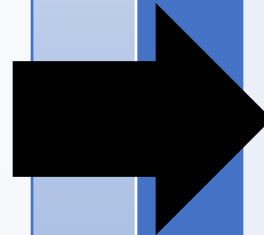
Pressing

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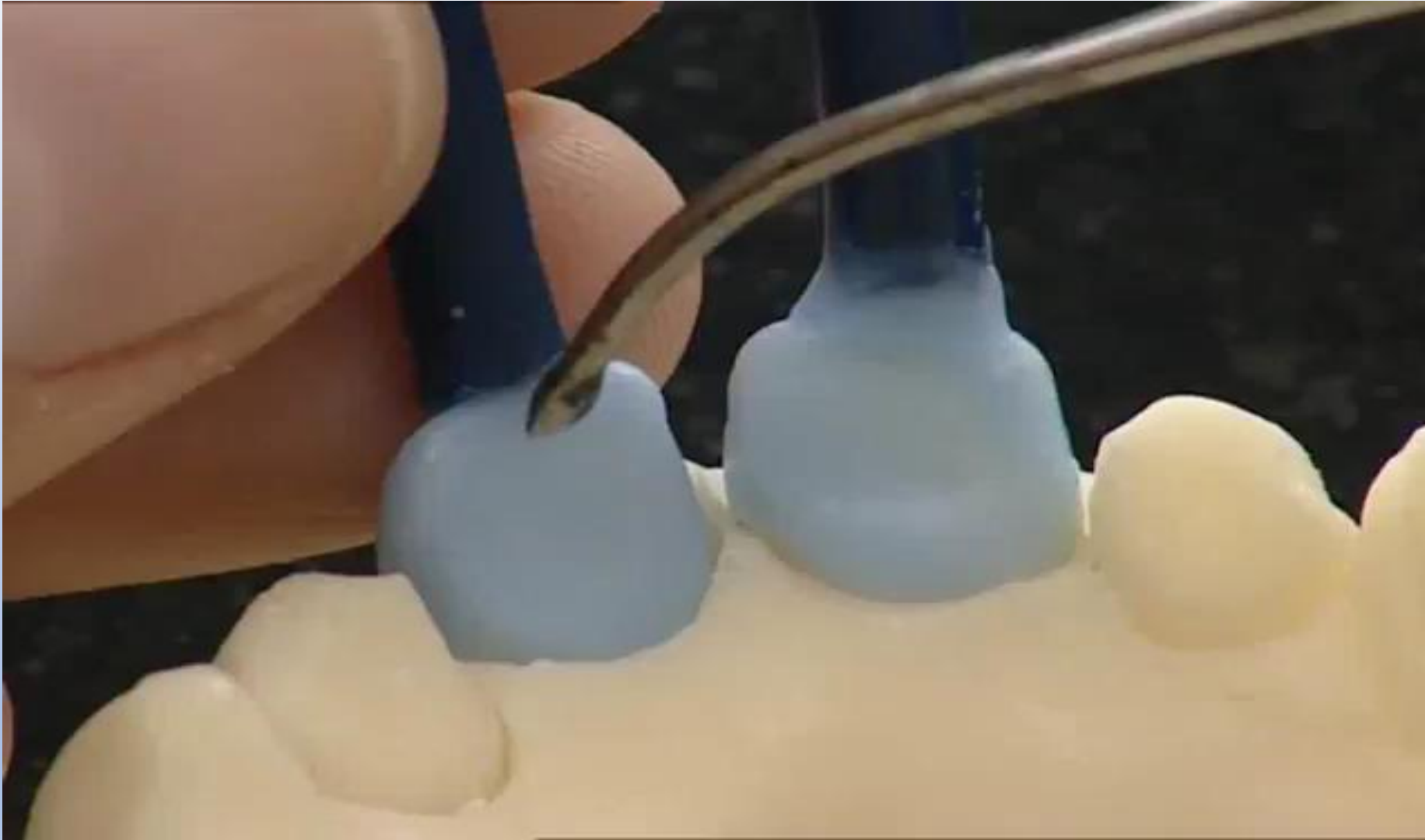
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D-Pressing techniques: are used to force a viscous mass of molten ceramic into a mold to get the desired final form, This process is similar to casting alloys , except that the ceramic, unlike the alloy, is too viscous to accelerate into the mold, and high pressure is used to move the ceramic.



Pressing is often employed today for making all-ceramic cores from glass-dominated or crystalline-dominated ceramics

Practical



Machining

E-Machining: Newer ceramics are machined into final form.

The replica of the tooth preparation is scanned into a computer, and then a highly sophisticated program drives fine machining tools to mill a ceramic block to the final form, usually in several minutes.

In the most advanced techniques, the shape of the tooth preparation is captured digitally (so-called digital impression), and these data are sent to the lab for milling



This technique is called Computer aided design & computer-aided manufacturing (CAD-CAM)



DENTAL CERAMICS

Processing methods



References

1- Dental materials foundations and applications 11th Edition

2- Craig's restorative dental materials 14th Edition.